Requirements

CO3 G10

Will Barlow
Harris Byrom
Corbin Hedger
Dmytro Kovalenko
Sharjil Munshi
Oliver Spence
Joshua Thomas
Leon Varghese

Introduction

To elicit requirements, we first analysed the product brief from the client to gain an idea of the project. This included an overview of the game with primary objectives, and a list of must-haves. We used this as a guide to form interview questions for the client to clarify any unknowns about the project and gain insight into key details that were not covered in the product brief. The interview was audio recorded so that we could later document the responses to each of the questions which were used to aid us in the formation of the requirements.

After some research into methods of presenting requirement specifications we decided to present our requirements using three tables:

- The first table is for user requirements which has a requirement ID, a description of the requirement, and a priority scaling from 'may' to 'shall'. Requirements with higher priorities will make up our initial focus when producing the actual project, whereas ones with lower priorities will be added if there is time and all high priority requirements have been met.
- The second table contains the functional system requirements which have a requirement ID, a description of the requirement, and the ID of the user requirement that it links to.
- The third table contains the non-functional system requirements which have the same columns as the second table but also has a fit criteria column which is a measurable definition of what it takes to satisfy the given requirement.

The IDs used to name our requirements are given meaningful names so that they can be better understood and identified quicker when referred to in different contexts outside of the requirement specification such as UML diagrams.

The requirement tables can be found on the following pages.

User Requirements

ID	Requirement	Priority
UR_GAME	Shall be a 2D maze game for any desktop OS.	Shall
UR_JAVA_17	Shall use Java 17.	Shall
UR_REASONABLE_HARDWARE	Should run well on any reasonable hardware and look good full screen on a normal screen size.	Should
UR_INTUITIVE_CONTROLS	Shall have intuitive controls.	Shall
UR_START_SCREEN	Should have a form of start/settings screen with access to a demo/tutorial.	Should
UR_PAUSABILITY	Should be able to pause at any time.	Shall
UR_END_SCREEN	Should have some form of end screen showing your final score.	Should
UR_SCORE_DURING_GAME	May be able to see your score during the game.	May
UR_FUN_GAME	Shall be fun.	Shall
UR_REPLAYABLE	May be replayable.	May
UR_MULTIPLE_ESCAPES	Should have multiple escape paths	Should
UR_UNI_LIFE_THEME	Shall have a theme of university life.	Shall
UR_SOLVABLE_IN_FIVE_MINS	Should be solvable in 4-5 mins for a new player.	Should
UR_SCORE_INP_FROM_TIME_AND_EVENTS	Score shall have input from time taken and other game elements.	Should
UR_ACCESSIBLE	Shall be accessible to the audience. For example by having different difficult options.	Shall
UR_MUSIC_OR_SOUND	May have music or sound effects.	May
UR_TOPDOWN_VIEW	Shall have a fixed top down view of the whole maze.	Shall
UR_RESPECTFUL	Shall be respectful of real people and places.	Shall
UR_BEGINNER_FRIENDLY	Should be beginner friendly.	Should
UR_TIME_LIMIT	Shall have a 5 minute time limit.	Shall
UR_NARRATIVE	Should have a consistent narrative.	Should
UR_OPEN_SOURCE	Shall use open-source libraries in accordance with their license terms.	Shall
UR_EVENTS	Shall have 5 visible events that will hinder from progressing, 3 visible that will benefit the player, and 3 hidden with fun interactions. Events may have variance in their placement.	Shall

Functional System Requirements

ID	Requirement	Reference
FR_JAVA_17	The system shall use Java 17.	UR_JAVA_17
FR_TWO_DIMENSION	The system shall be a 2D maze game.	UR_GAME
FR_DESKTOP	The system shall run on any desktop OS with a Java 17 runtime environment.	UR_GAME, UR_JAVA_17
FR_FULLSCREEN	The system shall allow the game to be played in fullscreen mode.	UR_REASONABLE_HARDWARE
FR_START_SCREEN	The system shall have a start screen with access to any applicable settings.	UR_START_SCREEN
FR_TUTORIAL	The system shall have a tutorial or guidance on how to play the game.	UR_START_SCREEN
FR_PAUSABILITY	The system shall allow the player to pause at any moment during gameplay.	UR_PAUSABILITY
FR_END_SCREEN	The system shall have a dedicated screen for when the game ends.	UR_END_SCREEN
FR_END_SCORE	The system shall display the end score	UR_END_SCREEN
FR_MUSIC_OR_SOUND	The system may be able to output sound.	UR_MUSIC_OR_SOUND
FR_TIME_LIMIT	The system shall have a 5 minute time limit.	UR_TIME_LIMIT
FR_EVENT_VARIANCE	The system may have variance in event placement.	UR_EVENTS
FR_SCORE_DURING_GAME	The system may display the score during the game	UR_SCORE_DURING_GAME
FR_MULTIPLE_ESCAPES	The system should have multiple escape paths	UR_MULTIPLE_ESCAPES
FR_UNI_LIFE_THEME	The system shall have a theme of university life	UR_UNI_LIFE_THEME
FR_SOLVABLE_IN_FIVE_MINS	The system should be solvable in 4-5mins from a player	UR_SOLVABLE_IN_FIVE_MINUTES
FR_INP_FROM_TIME_AND_EVENTS	The system shall have input from time taken and other elements	UR_INP_FROM_TIME_AND_EVENTS
FR_NEGATIVE_EVENTS	The system shall have at least 5 visible events that will hinder from progressing	UR_EVENTS
FR_POSITIVE_EVENTS	The system shall have at least 3 visible events that will benefit the player	UR_EVENTS
FR_HIDDEN_EVENTS	The system shall have at least 3 events with fun interactions	UR_EVENTS
FR_OPEN_SOURCE	The system shall use open source libraries in accordance with their terms of use	UR_OPEN_SOURCE

FR_TOPDOWN_VIEW	The game shall have a fixed topdown view of the maze.	UR_TOPDOWN_VIEW
FR_DIFFICULTY_OPTIONS	The game may have different difficulty options.	UR_ACCESSIBLE
FR_SCORE_INP_FROM_TIME_AND_EVENTS	The score should be affected by both time taken, and individual events.	UR_SCORE_INP_FROM_TIME_AND_EVENTS

Non-Functional System Requirements

ID	Requirement	Reference	Fit Criteria
NFR_PERFORMANCE	The system shall run well on any modern desktop computer, running a modern OS and a Java 17 runtime environment.	UR_REASONABLE_HARDWARE, UR_JAVA_17	Should run at our target framerate 90% of the time on any system running the latest version of any currently supported Windows, MacOS, GNU/Linux OS built within the last 5 years.
NFR_FULLSCREEN_GOOD	The system shall look good in full screen.	UR_REASONABLE_HARDWARE	Fullscreen mode should run at the native screen resolution for at least 90% of surveyed users.
NFR_INTUITIVE_CONTROLS	The game's controls should be intuitive.	UR_INTUITIVE_CONTROLS	At least 90% of surveyed users should find the controls to be "quite intuitive" or better.
NFR_RESPECTFUL	The game shall be respectful of real people and places.	UR_RESPECTFUL	At least 90% of surveyed users should find the game to be "respectful" or better.
NFR_NARRATIVE	The game should have a consistent narrative.	UR_NARRATIVE	At least 90% of surveyed users should find the narrative to be "consistent" or better.
NFR_FUN_GAME	The system shall be fun.	UR_FUN_GAME	At least 90% of surveyed users should find the game to be "fun" or better.
NFR_ACCESSIBLE	The system shall be accessible to all users, including beginners.	UR_ACCESSIBLE, UR_BEGINNER_FRIENDLY	At least 90% of surveyed users should find the game to be "quite accessible" or better.
NFR_REPLAYABLE	The system may be replayable.	UR_REPLAYABLE	At least 90% of surveyed users should find the game to be "quite replayable" or better.